OFFICIAL BIDDING DOCUMENT

Telecommunications Technology Solutions Inc. (TelTech)

Corporate Tender Secretariat · Procurement & Network Engineering Division

RFP-TELTECH-FWA-2026-001 / APPENDIX D

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Site Survey, RF Planning & Civil/Power Engineering Guidelines

TowerCo Co-Location Standard, Greenfield Siting & Link Budget Engineering Standards

DOCUMENT NUMBER	AUTHOR	TARGET SPECTRUM	BIDDING STATUS
RFP-TELTECH-FWA-2026-001	OLIVER TUNGOL, PMP, PECE	Band n50 (1.5 GHz, 60 MHz TDD)	Official Request for Proposal

1. ENGINEERING PRINCIPLES & SITING ARCHITECTURE

1.1 Siting Philosophy & Mandatory 3-Tier Sourcing Hierarchy

The primary engineering objective of the 5G FWA rollout is to maximize household population coverage within high-density suburban and rural barangay settlement clusters while minimizing civil capital expenditures and permitting lead times. To achieve rapid 1,000-site Phase 1 execution within 12 months, the Contractor shall execute site delivery strictly following the **Mandatory 3-Tier Infrastructure Sourcing Hierarchy**:

1. **Tier 1 (Mandatory Baseline — Existing TowerCo Co-Location):** Field teams must prioritize co-location on existing operational towers (PTCI/MIDC, EdgePoint, FTAP Globe, FTAP Smart, LDIC, Phil-Tower) located within the designated search ring. The Operator leverages an audited master database of **13,810 nationwide TowerCo registry records (encompassing 13,799 unique Universal Tower IDs and 13,629 distinct geographic tower coordinates across PTCI/MIDC, EdgePoint, LDIC, FTAP Globe, and FTAP Smart)**. Co-location reduces Ready-for-Installation (RFI) lead times from 120–180 days down to 15–30 days while eliminating private land titling and environmental permitting complexities.
2. **Tier 2 (Secondary — TowerCo Build-to-Suit / B2S):** If all existing towers within the search ring exceed structural loading thresholds (EPA limits) or fail RF line-of-sight requirements, the vendor shall engage a licensed Independent TowerCo to execute a Build-to-Suit (B2S) under TelTech's Master Lease Agreement (MLA).
3. **Tier 3 (Restricted Exception — Operator Greenfield Own-Build):** Direct operator-constructed greenfield monopoles are strictly limited to search rings where no feasible or available TowerCo infrastructure exists after documented technical and field checks. Qualification for Tier 3 Greenfield Own-Build operates strictly under two mutually exclusive pathways:
   - **Pathway A (Zero Infrastructure / Siting Void):** Applicable where the audited master tower database and joint field survey confirm the total absence of any operational tower within the primary search ring (500m urban / 1,000m rural) AND within the secondary candidate zone (≤ 2.0 km line-of-sight). Substantiation: Certified Absence of Tower Infrastructure in the approved joint TSSR, verified against the audited master database and 360° drone/visual survey.
   - **Pathway B (Technical / Structural Exhaustion of Existing Infrastructure):** Applicable where one or more operational towers exist within the search ring or within 2.0 km, but co-location is physically or structurally unfeasible due to: (i) structural over-utilization / EPA overload where the TowerCo confirms the tower cannot support additional 3x 8T8R AAU mounts and foundation reinforcement is unviable; (ii) physical compound/aperture exhaustion (zero ground space for outdoor DC power pad or no vertical mounting aperture at required AGL); or (iii) unresolvable terrain/foliage clutter blocking optical Fresnel zone line-of-sight into the target settlement cluster. Substantiation: Formal written **Structural Denial / Space Unavailability Letter(s)** issued by the operating TowerCo(s) under the MLA framework (a single denial letter if only 1 TowerCo operates

in the ring; minimum 2 letters if multiple TowerCos operate within the ring), accompanied by TSSR RF simulation proving coverage blockage.

4. **Phase 1 Quantity Reconciliation & Survey Exceptions:** The evaluated commercial baseline across all RFP documents (Master Tender Section 2.1, BoQ Table 6, and Bulletin 01) is formally established at **897 Co-Location Sites (89.7%) and 103 Greenfield Sites (10.3%)**. In Document 07, the preliminary spatial proximity screening against the audited tower database classifies 893 Co-location Candidates (89.3%) within ≤ 2.0 km (551 Class T1 ≤ 0.5 km; 255 Class T2 0.5–1.0 km; 87 Class T3 1.0–2.0 km) and 107 Greenfield Monopole Candidates (10.7%) at Class T4 (> 2.0 km) nationwide (and 4,517 co-location / 483 greenfield across all 5,000 sites). Co-location candidate references located between 1.0 km and 2.0 km from the demand centroid (including 174 Phase 1 candidate references) are formally designated **Survey Exception Candidates**: during Milestone 1 TSSR, field teams evaluate whether clear line-of-sight and link budget (RSRP ≥ -105 dBm) can be achieved from the out-of-ring tower; if confirmed, the site executes as Tier 1 Co-Location; if obstructed or unfeasible, the candidate converts under Pathway A or B to Tier 2 B2S or Tier 3 Greenfield at the nominal demand centroid.
5. **Commercial Valuation & Siting True-Up Mechanism:** While tender evaluation and contract award are governed by the lump-sum pricing of the 897 Co-Location / 103 Greenfield evaluated baseline (BoQ Table 6), actual contractor compensation will be trued up at Milestone 1 (TSSR Approval). If an evaluated co-location site converts to Tier 2 B2S or Tier 3 Greenfield (or vice-versa), contract adjustments shall be calculated strictly using Document 06 Table 5B (Siting Conversion & Milestone 1 True-Up Rate Schedule) via an administrative contract variation order:
   - **Tier 1 Co-Location Baseline (`LOT5-COLO`):** Rate = Item CIV-SVC-03 ($0.00 variation differential).
   - **Tier 2 TowerCo Build-to-Suit (`LOT5-B2S`):** Under the Operator MLA, the TowerCo finances, permits, and constructs the tower and compound turnkey; Operator civil scope is identical to co-location (`CIV-SVC-03`). Rate = `LOT5-B2S` = `CIV-SVC-03`. Variation differential vs. Co-Location = **$0.00 / site** (`LOT5-B2S − LOT5-COLO = $0.00`).
   - **Tier 3 Greenfield 30m Monopole (`LOT5-GF30`):** Compound foundation, 8x8m fence/gravel, grounding grid, 30m monopole, and SAQ permits (`CIV-SVC-02 + CIV-SVC-04 + CIV-SVC-05 + CIV-SVC-06`). Net contractual addition per converted co-location site: **`ΔCost = LOT5-GF30 − LOT5-COLO = (CIV-SVC-02 + CIV-SVC-04 + CIV-SVC-05 + CIV-SVC-06) − CIV-SVC-03`**. Reverse conversion from GF30 to Co-Lo issues an equivalent credit.
   - **Tier 3 Greenfield 40m Monopole (`LOT5-GF40`):** Compound foundation, 8x8m fence/gravel, grounding grid, 40m monopole, and SAQ permits (`CIV-SVC-02 + CIV-SVC-04 + CIV-SVC-05 + CIV-SVC-07`). Net contractual addition per converted co-location site: **`ΔCost = LOT5-GF40 − LOT5-COLO = (CIV-SVC-02 + CIV-SVC-04 + CIV-SVC-05 + CIV-SVC-07) − CIV-SVC-03`**. Reverse conversion from GF40 to Co-Lo issues an equivalent credit.

1.2 Search Ring Specification & Demand Centroid

For each nominal site location assigned in Document 07 (Candidate Site List):

- **Urban & Suburban Search Ring:** Defined strictly as a circle with a radius of **500 meters** centered on the official nominal barangay demand centroid.
- **Rural & Remote Search Ring:** Defined strictly as a circle with a radius of **1,000 meters** centered on the official nominal barangay demand centroid.
- **Search Ring vs. RF Coverage Radius Distinction:** The search ring defines the geographic hunting boundary for candidate site sourcing and infrastructure identification. It is completely distinct from the physical RF cell coverage radius (nominally 2.5–3.2 km in rural terrain and 1.0–1.5 km in urban clutter per Section 3.2).
- **Target Coverage Guarantee:** Sited candidate coordinates within the search ring must achieve unobstructed Line-of-Sight (LOS) and target RSRP coverage (≥ -95 dBm) across ≥ 90% of the inhabited settlement zone.

1.3 Mandatory Three-Candidate Rule (Candidates A, B, C) & Commercial Authority

For every search ring, the Turnkey Contractor / Site Acquisition Vendor must survey, evaluate, and submit **three (3) ranked candidates** in the Technical Site Survey Report (TSSR):

- **Candidate A (BEST — Primary Recommended):** Optimal TowerCo co-location asset (or prime site) with immediate structural capacity at ≥ 30m AGL, available -48V DC busbar or commercial AC transformer headroom, nearest

fiber node, and fastest projected RFI (≤ 21 calendar days).

- **Candidate B (BETTER — Secondary Target / Backup 1):** Fully compliant alternative TowerCo structure or site within the search ring requiring minor solvable modifications (e.g., dedicated AC sub-meter installation, mount bracket extensions, or secondary fiber lead-in route).
- **Candidate C (GOOD — Contingency / Fallback):** Third viable TowerCo asset or TowerCo Build-to-Suit option, ensuring the search ring is not stalled if Candidates A and B encounter unforeseen structural or commercial deadlocks.
- **Exception for Rural Sites:** Where fewer than three viable structures or constructible parcels exist within a rural 1,000m search ring, the Contractor must submit formal documented justification in the TSSR.
- **Commercial Negotiation & Lease Execution Authority:** In accordance with Bid Bulletin No. 1 (Item 17), all commercial lease negotiations, space reservations, and Master Lease Agreement (MLA) Service Orders with TowerCos are **strictly the exclusive responsibility of the Operator (TelTech/Comclark)**.
- **Contractor Structural Audit vs. TowerCo Analysis:** The Contractor's scope is strictly technical on-site auditing (recording physical dimensions, mounting heights, existing antennas, and compiling the proposed equipment loading profile). Formal structural engineering analysis, wind stress modeling, foundation capacity verification, and PE sign-off are executed by the TowerCo under the Operator's Master Lease Agreement (MLA).
- **14-Day Technical Escalation Protocol:** If Candidate A fails structural loading verification or if Operator-TowerCo commercial SAA processing encounters an impasse exceeding **14 calendar days**, the Operator shall instruct the Contractor to immediately activate field engineering and TSSR finalization for **Candidate B** to prevent project schedule slippage.

1.4 Co-Location Candidate Evaluation Criteria Matrix (100-Point Scoring)

- **Space & Structural Capacity (20%):** Antenna mounting space at ≥ 30m to 45m AGL, wind load headroom for 3x n50 antennas + 3x 8T8R RRUs (≤ 45 kg each), and ≥ 2.0m × 2.0m compound ground space for outdoor rectifier/battery cabinets.
- **Access Road & Compound Security (15%):** All-weather paved/concrete road access, established TowerCo security fencing/guarding, and 24/7 unobstructed Operations & Maintenance (O&M) ingress/egress.
- **RF Line-of-Sight & Propagation (25%):** Dominant ground elevation AMSL, clear first Fresnel zone into residential settlement, zero close-in clutter shielding within 200m of sector azimuths (0°, 120°, 240°).
- **Backhaul Transport Proximity (15%):** Proximity to Converge/TelTech fiber distribution closure (Class A ≤ 0.5km, Class B ≤ 1.0km), existing TowerCo cable ladder/conduit entry, or unobstructed northern sky view (0° – 80° elevation arc) for Starlink satellite backhaul.
- **Power Availability & Utility Infrastructure (15%):** Commercial utility grid transformer capacity, available -48V DC busbar tap (≥ 63A branch breaker dedicated for radio equipment) or space for dedicated AC service entrance sub-meter, and shared backup genset/battery support. Total site power plant sizing (12.0 kW N+1) is governed by Lot 2.
- **Lease & Commercial Feasibility (10%):** Feasibility of fast-track Service Order execution under existing TowerCo Master Lease Agreement (MLA), monthly recurring fee within Operator baseline budget, and projected RFI timeline ≤ 30 calendar days.

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2. DETAILED TECHNICAL SITE SURVEY REPORT (TSSR) REQUIREMENTS

For every nominal site candidate, the Contractor shall complete and submit a comprehensive TSSR conforming to the following structure:

2.1 General Site Identification & Geography

- Site Name, Candidate ID, Official PSGC Code, Region, Province, Municipality, and Barangay.
- Precision WGS84 GPS Coordinates (Latitude, Longitude recorded with >= 6 decimal places via dual-frequency GPS receiver).

- Ground Elevation Above Mean Sea Level (AMSL in meters).
- Access Road Assessment: Road surface type (concrete, asphalt, macadam, dirt trail), bridge weight limits, and 4x4 hauling accessibility during monsoon seasons.

2.2 Tower Structure & Loading Audit (Co-Location Sites)

- Operating TowerCo Entity (PTCI/MIDC, FTAP Globe, EdgePoint, FTAP Smart, LDIC, FTAP BTS).
- Tower Type (Self-Supporting 3-Leg/4-Leg Tower, Monopole, Guyed Mast, Rooftop Tower).
- Tower Height Above Ground Level (AGL in meters).
- Structural Space Audit: Proposed mounting height for 3x Band n50 antennas (recommended centerline: 30m to 45m AGL).
- Structural Loading Verification: Confirmation that existing tower members and foundation can accommodate additional 3x 8T8R RRUs/AAUs (< 45 kg each), antenna mounts, and nominal design wind speed of 250 km/h (plus applicable NSCP 2015 ultimate wind zone rating up to 315 km/h).
- Panoramic Drone Photographic Survey: High-resolution 360° continuous panoramic video and 12-cardinal-point still photos at proposed antenna centerline height to verify optical line-of-sight across populated settlements. *Conditional Permitting Waiver:* Where drone flight is legally prohibited or restricted by CAAP regulations or military airfields, OR where local municipal/flight permits cannot be secured within fourteen (14) calendar days, high-magnification telephoto mast photography from multiple accessible ground vantage points combined with verified TowerCo structural drawings is accepted subject to Operator approval per Bid Bulletin No. 1.

2.3 Power Infrastructure & Utility Audit

- Existing Commercial AC Utility Power: Electric Cooperative / Distribution Utility name (e.g., MERALCO, PELCO, BENECO, CAPELCO, ILPI).
- Service Entrance Capacity: Transformer rating (kVA), service voltage (230V / 400V, 60 Hz), breaker rating, and phase balance.
- TowerCo DC Power Availability: Sourcing from TowerCo DC busbar (-48V DC, ≥ 63A dedicated branch circuit breaker for radio equipment ~3.0 kW) vs. installing Operator-dedicated rectifier subrack within the compound (total site power plant sized for 12.0 kW N+1 per Lot 2).
- Sub-metering feasibility and cable tray routing for power cables.
- Grounding System Audit: Measurement of existing earth ground resistance using a 3-point fall-of-potential earth tester. Earth resistance must be <= 5.0 Ohms.

2.4 Backhaul Transmission Path Survey

- Audited Converge Fiber Network Match (subject to verification on the ground): Distance to nearest confirmed node candidate (Class A <= 0.5 km, Class B <= 1.0 km) or line-tap closure (Class C <= 0.5 km).
- Cable Entry Facility: Manhole, handhole, or aerial pole lead-in routing into tower shelter/compound.
- For Remote Satellite Sited Candidates (Starlink Business): Satellite antenna mounting location (unobstructed sky view with 0° to 80° northern elevation arc for Starlink constellation tracking).

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3. RF PROPAGATION & CELL DIMENSIONING GUIDELINES

3.1 Dual-Tier Spectrum Architecture & Operating Bandwidth (Band n50)

In strict alignment with **Bid Bulletin No. 1 (Section 2.1)** and NTC frequency authorizations, the 5G FWA network operates on a dual-tier spectrum deployment model governed by regional utility franchise boundaries:

1. **Metro Manila (NCR) & Meralco Franchise Footprint:**
   - **Operating Frequency Range: 1457.0 – 1517.0 MHz (60.0 MHz continuous TDD carrier)**.
   - **Center Frequency: 1487.0 MHz**.

- **Transmission Bandwidth: 162 Resource Blocks (RBs)** at 30 kHz Subcarrier Spacing (SCS).
- *Spectrum Boundary Condition:* In Metro Manila (NCR) and non-NCR municipalities within the Meralco electric distribution franchise (parts of Bulacan, Cavite, Rizal, Laguna, Batangas, Quezon, and Pampanga) where the 1437.0 – 1457.0 MHz sub-band is assigned to the utility corporation, the 1437.0 – 1457.0 MHz block is excluded from Operator transmissions. Digital filtering within the RRU transceiver enforces sharp attenuation at 1457.0 MHz.

2. **Rest of the Country (Provincial / Regional / Non-Meralco Areas):**
   - **Operating Frequency Range: 1437.0 – 1517.0 MHz (80.0 MHz continuous TDD carrier — Desired Baseline)**.
   - **Center Frequency: 1477.0 MHz**.
   - **Transmission Bandwidth: 217 Resource Blocks (RBs)** at 30 kHz SCS.
   - *Capacity & Throughput Advantage:* In all provincial municipalities and rural barangays outside the Meralco territory, the full 80 MHz channel is activated, unlocking +34% additional physical resource blocks and driving aggregate sector downlink capacity to 650 – 720 Mbps per site.
3. **Mandatory Hardware Tuning & Filtering Specifications:**
   - **Native Hardware IBW:** All proposed 5G NR Baseband Units (BBU) and Remote Radio Units (RRU) / Active Antenna Units (AAU) must natively support the full 3GPP Band n50 tuning range (**1432.0 – 1517.0 MHz**) in hardware.
   - **Software-Defined Bandwidth:** Transceivers must support dynamic software reconfiguration across 40 MHz, 50 MHz, 60 MHz, and 80 MHz without physical component replacement or tower climbs.
   - **Adjacent Satellite Protection:** Transceivers must comply with **ITU-R M.2159-0 / WRC-15 Resolution 223** to protect adjacent Maritime Mobile Satellite Service (MSS, 1518.0 – 1525.0 MHz). Unwanted out-of-band emissions at 1517.0 MHz must not exceed -30 dBm/MHz at the antenna port (3GPP TS 38.104 Category B).

3.2 Link Budget Parameters (Band n50, 1.5 GHz)

PARAMETER	UNIT	60 MHZ MODE (NCR / MERALCO)	80 MHZ MODE (PROVINCIAL / RURAL)	TECHNICAL NOTES & MODELING BASIS
Operating Frequency	MHz	1487.0 (Center)	1477.0 (Center)	Band n50 TDD
Channel Bandwidth	MHz	60.0 MHz	**80.0 MHz**	162 RBs (60 MHz) vs. **217 RBs (80 MHz)**
Subcarrier Spacing (SCS)	kHz	30.0	30.0	5G NR Numerology μ=1
BTS Maximum Conducted Power	dBm / W	55.0 dBm (320 W)	55.0 dBm (320 W)	8 x 40W RRU configuration
BTS Power per Resource Block	dBm	32.9 dBm	**31.6 dBm**	Uniform power allocation across active RBs
BTS Antenna Gain	dBi	17.5 dBi	17.5 dBi	8-port cross-pol macro antenna
BTS Cable / Jumper Loss	dB	1.0 dB	1.0 dB	Low-loss jumper interface
CPE Maximum Transmit Power	dBm / W	23.0 dBm (200 mW)	23.0 dBm (200 mW)	Power Class 3 UE (or Class 2 26 dBm)
CPE Antenna Gain	dBi	9.0 dBi	9.0 dBi	Baseline model (6 dBi indoor / 12 dBi outdoor)
CPE Body / Indoor Penetration Loss	dB	8.0 dB	8.0 dB	Concrete hollow block & glass attenuation
Foliage / Clutter Margin	dB	6.0 dB	6.0 dB	Typical rural tropical tree canopy

PARAMETER	UNIT	60 MHZ MODE (NCR / MERALCO)	80 MHZ MODE (PROVINCIAL / RURAL)	TECHNICAL NOTES & MODELING BASIS
Shadowing Margin (Log-Normal)	dB	7.0 dB	7.0 dB	90% edge cell coverage reliability
Thermal Noise Density	dBm/Hz	-174.0	-174.0	kT @ 290K
Receiver Noise Figure	dB	6.0 dB (CPE) / 3.5 dB (gNB)	6.0 dB (CPE) / 3.5 dB (gNB)	Active RF front-end
Target SINR for Min Service	dB	-1.0 dB (QPSK 1/3)	-1.0 dB (QPSK 1/3)	Cell-edge SLA threshold (+10 dB for 64QAM)
Maximum Allowable Path Loss (MAPL)	**dB**	**139.8 dB**	**138.5 dB**	**Uplink coverage limiting (Δ ≈ 1.3 dB for 80 MHz)**
Calculated Reliable Cell Radius	**km**	**1.0 – 1.5 km (Urban/Suburban Clutter — 3GPP UMa)**	**2.5 – 3.2 km (Rural Baseline / 1.5 – 2.0 km Vendor Benchmark)**	**Distinct from search ring (500m Urban / 1,000m Rural)**
Peak Aggregate Sector DL Throughput	**Mbps**	**~480 – 520 Mbps**	**~650 – 720 Mbps**	**Multi-user MIMO, 256QAM downlink**

3.3 Nominal Sector Azimuth & Tilt Strategy

- **Sector Count:** 3 sectors per macro site with 120° nominal separation (0°, 120°, 240°). Azimuths must be tailored during TSSR to track actual barangay population concentrations along transport corridors and coastal plains.
- **Electrical Remote Downtilt (eRET):** Antennas must feature AISG 2.0 remote tilt adjustable between 2° and 12° from the central OSS. Mechanical tilt shall only be applied when required to clear close-in obstructions (>15°).

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4. CIVIL WORKS & STRUCTURAL ENGINEERING (CLASS T4 GREENFIELD & TOWERCO BTS SITES)

For sites requiring self-constructed greenfield towers or TowerCo Build-to-Suit (B2S) structures, the Contractor shall adhere to the following civil and structural standards conforming to the **National Structural Code of the Philippines (NSCP 2015 7th Edition, Section 207 Wind Loads)** and **TIA-222-G/H**:

4.1 Philippine Basic Wind Speed Regimes & Wind Zone Mapping

Structural design calculations and wind force determinations must strictly comply with the **National Structural Code of the Philippines (NSCP 2015 7th Edition, Section 207)** and **TIA-222-G/H**. Structure classification: Essential Communications Facility / Risk Category IV (Importance Factor I_W = 1.15). The nominal basic design wind speed for structural strength and foundation stability calculations is **250 km/h (Nominal / ASD basic wind speed per NSCP 2015 Category IV)**. Ultimate / Survivable wind speeds conform strictly to NSCP 2015 Wind Zones:

- **Wind Zone 1 (Ultimate Wind Speed V = 315 km/h / 87.5 m/s):** Covers the high-typhoon corridor along the eastern seaboard (Batanes, Cagayan, Isabela, Aurora, eastern Quezon coast and Polillo Islands, Bicol Region, Samar, Leyte, Dinagat Islands, Surigao del Norte/Sur, Davao Oriental).
- **Wind Zone 2 (Ultimate Wind Speed V = 270 km/h / 75.0 m/s):** Covers Central and Western Luzon, National Capital Region (NCR), Central Luzon (Pampanga, Bulacan, Tarlac, Bataan, Zambales, Nueva Ecija), CALABARZON (western Quezon, Laguna, Batangas, Cavite, Rizal), MIMAROPA (Occidental/Oriental Mindoro, Marinduque, Romblon), and Western/Central Visayas (Iloilo, Cebu, Bohol, Negros).
- **Wind Zone 3 (Ultimate Wind Speed V = 250 km/h / 69.4 m/s Minimum Project Floor):** Covers low-risk regions including Palawan, Zamboanga Peninsula, and southern inland Mindanao. While NSCP Section 207 maps Zone 3 basic wind speed at 200 km/h, the Project enforces 250 km/h minimum ultimate survivability nationwide as a mandatory contractual quality floor.

4.2 Structural Limit States & Deflection Criteria

- **Survivable Wind Speed ($V_{survival}$ — Ultimate Limit State / LRFD):** Designed for **250 km/h to 315 km/h** (250 km/h national project floor; 270 km/h in Zone 2; 315 km/h in Zone 1) 3-second gust wind speed under Ultimate Limit State with full equipment loading (3x n50 antennas, 3x RRUs, satellite dish, and mounting steelwork). Zero plastic collapse or structural overturning.
- **Operational Wind Speed ($V_{operational}$ — Serviceability Limit State / SLS):** Designed for **140 km/h to 160 km/h (40.0 m/s to 44.4 m/s)** continuous operational wind speed (1.0 load factor per TIA-222). Structure maintains rigid RF alignment, ensuring uninterrupted 5G beamforming, MIMO rank-4 transmission, and backhaul link availability (≥ 99.995%).
- **Allowable Twist (θ_{twist} / Torsional Rotation):** ≤ ±0.5° (30 arcmin) at 30m AGL for narrow-beam / microwave dish antennas; ≤ ±1.0° (60 arcmin) at 30m AGL for 5G Band n50 macro sectors.
- **Allowable Sway / Tilt (θ_{sway} / Angular Deflection):** ≤ ±1.0° (60 arcmin) at top antenna mounting centerline (30.0m AGL).
- **Allowable Verticality (Tower Plumbness / Alignment):** ≤ 1 : 500 (0.20% of Tower Height H) throughout the full mast height under dead load and still air. For a 30.0m monopole, maximum horizontal deviation $\Delta \leq 60$ mm (for 40.0m monopole, $\Delta \leq 80$ mm). Tolerance between consecutive slip-joints: ≤ 1 : 1000 (0.10%).
- **Horizontal Service Deflection (Δ_{top}):** ≤ H / 100 (300 mm maximum at 30.0m AGL).

4.3 Monopole Tower & Material Specification

- **Tower Structure:** 30-meter or 40-meter polygonal tubular steel monopole constructed in 3 to 4 slip-joint sections with minimum 1.5 × inside diameter overlap.
- **Structural Steel:** ASTM A572 Grade 50 (F_y = 345 MPa / 50 ksi, F_u = 450 MPa).
- **Corrosion Protection:** Hot-dip galvanized per ASTM A123 / ISO 1461 with minimum zinc coating thickness ≥ 85 µm (610 g/m²).
- **Fasteners & Anchors:** Anchor bolts ASTM F1554 Grade 55/105 with double heavy-hex nuts; structural bolts ASTM A325 / ISO 898-1 Class 8.8 hot-dip galvanized.

4.4 Compound Civil Works & Foundation

- **Compound Footprint:** Standard 8.0m × 8.0m (64 m²) gravel-topped compound with geotextile weed barrier.
- **Perimeter Fencing:** 2.4-meter high cyclone wire mesh fence topped with 3 strands of galvanized concertina razor wire.
- **Equipment Pad:** Reinforced concrete equipment plinth (Class A concrete, 28-day compressive strength $f'_c \geq 28$ MPa / 4,000 psi, ASTM A615 Grade 60 rebar) for outdoor power/battery cabinets.
- **Foundation:** Reinforced concrete pad and chimney mat foundation or bored piles engineered with Safety Factor ≥ 2.0 against soil bearing failure and ≥ 1.5 against overturning/sliding.

4.5 Lightning Protection & Earthing Grid

- Top-mounted Franklin lightning air terminal extending ≥ 2.0m above uppermost antenna.
- Down-conductor: 70mm² bare stranded copper cable routed externally and clamped every 1.0m to tower shaft.
- Ground ring: Continuous buried bare copper loop around tower base and compound with copper-clad steel ground rods (19mm dia × 3.0m length) exothermically welded (Cadweld) at all junctions, achieving earth ground resistance ≤ 5.0 Ohms.

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5. SINGLE SITE VERIFICATION (SSV) & CLUSTER ACCEPTANCE CRITERIA

Before any site is granted Commercial On-Air Acceptance, the Contractor must complete and submit a signed SSV Report verifying:

1. **Stationary Functional Tests:**
* Sector Verification: Physical azimuth and tilt match TSSR approval within +/- 2°.
* RF Transmit Power: Conducted power on all 8 transmit paths within +/- 0.5 dB of nominal.
* VSWR / Return Loss: Return loss >= 15 dB across all antenna feeder lines.
* Max Throughput Test: Single CPE in line-of-sight achieving >= 300 Mbps DL and >= 50 Mbps UL (aggregate sector peak >= 450 Mbps DL / >= 60 Mbps UL verified under multi-CPE or loaded test).
2. **Short Drive Test & Static Multi-Point Test:**
* 5 stationary measurement points per sector (Near: 500m, Mid: 1.5 km, Far/Edge: 3.0 km).
* RSRP >= -95 dBm across >= 90% of surveyed barangay inhabited zone.
* Ping RTT Acceptance: For terrestrial fiber-fed sites, Ping RTT ≤ 18 ms target (≤ 30 ms maximum contractual limit) to local UPF IP address; for Starlink satellite-fed remote sites, Ping RTT ≤ 60 ms target (≤ 80 ms maximum contractual limit) reflecting LEO satellite propagation latency per Bid Bulletin No. 1; packet loss ≤ 0.10% over 1,000 continuous pings.
* Inter-sector handover execution without drop during transition.

(End of Appendix D — Site Survey, RF Planning & Civil Engineering Guidelines)
